

Dataset1 Ablation Report

Comparing the formal 5-feature baseline with the formal 6-feature run that adds CI2 dose, including performance metrics and SHAP interpretation.

Generated on 2026-03-31

1. Study setup

Design choices, dataset footprint, and model families used in both formal runs.

Scope

This report compares two formal dataset1 experiments: a 5-feature baseline (pH, UV254, temperature, TOC, bromide) and a 6-feature variant that adds Cl2 dose. Both runs use the same three targets (THM4, DBCM, BDCM), the same four model families (Random Forest, XGBoost, MLP, KAN), and the same per-target training paradigm.

Data and validation

- Raw dataset rows: 514 with a predefined split of 406 train and 108 test rows.
- After drop-missing complete-case filtering, the 5-feature run used 488 rows (382 train / 106 test).
- After drop-missing complete-case filtering, the 6-feature + Cl2 run used 488 rows (382 train / 106 test).
- Formal tuning used 5-fold cross-validation with a stability penalty of 0.18.
- Global neural training settings: max_epochs = 1200 and patience = 100.

Experiment matrix

Run	Features	Targets	Models
Formal 5-feature	pH, UV254, Temp, TOC, Bromide	THM4, DBCM, BDCM	RF, XGB, MLP, KAN
Formal 6-feature + Cl2	pH, UV254, Temp, TOC, Bromide, Cl2 dose	THM4, DBCM, BDCM	RF, XGB, MLP, KAN

2. Macro performance

Full-model macro metrics across the two formal ablation rounds.

Adding CI2 improved macro performance for Random Forest, XGBoost, and MLP. KAN did not benefit from the new feature in this formal run.

Model	5f RMSE	5f MAE	5f R2	6f RMSE	6f MAE	6f R2	Delta RMSE	Delta R2
Random Forest	23.9480	17.7539	0.6561	18.8186	13.7864	0.7335	-5.1294	+0.0774
XGBoost	27.7906	19.7944	0.6168	21.8283	14.8577	0.7338	-5.9623	+0.1170
MLP	26.1434	19.5004	0.6340	19.9756	14.9304	0.7607	-6.1678	+0.1267
KAN	27.5675	20.9970	0.5413	28.2994	18.3030	0.5409	+0.7319	-0.0004

Macro takeaways

- Random Forest: macro RMSE 23.95 to 18.82, macro R2 0.6561 to 0.7335
- XGBoost: macro RMSE 27.79 to 21.83, macro R2 0.6168 to 0.7338
- MLP: macro RMSE 26.14 to 19.98, macro R2 0.6340 to 0.7607
- KAN: macro RMSE 27.57 to 28.30, macro R2 0.5413 to 0.5409
- Best macro RMSE in the 6-feature run: Random Forest (18.82).
- Best macro R2 in the 6-feature run: MLP (0.7607).

3. Target-level results - THM4

Detailed RMSE, MAE, and R2 values for every model in both ablation rounds.

Model	5f RMSE	5f MAE	5f R2	6f RMSE	6f MAE	6f R2	Delta RMSE	Delta R2
Random Forest	55.4348	40.8749	0.6812	41.4572	29.8655	0.8217	-13.9776	+0.1405
XGBoost	66.5437	46.6447	0.5406	50.3912	33.4211	0.7365	-16.1525	+0.1960
MLP	61.6462	46.4169	0.6057	46.1545	33.6113	0.7790	-15.4917	+0.1733
KAN	63.8378	49.1624	0.5772	66.0904	42.3469	0.5468	+2.2527	-0.0304

Interpretation

- Best RMSE in the 5-feature baseline: Random Forest (55.43).
- Best RMSE in the 6-feature + CI2 run: Random Forest (41.46).
- The winning model family did not change for THM4.

3. Target-level results - DBCM

Detailed RMSE, MAE, and R2 values for every model in both ablation rounds.

Model	5f RMSE	5f MAE	5f R2	6f RMSE	6f MAE	6f R2	Delta RMSE	Delta R2
Random Forest	6.4705	4.7697	0.5051	6.1615	4.6024	0.5513	-0.3090	+0.0461
XGBoost	6.0639	4.6379	0.5654	5.1951	3.7411	0.6810	-0.8688	+0.1156
MLP	6.2209	4.2398	0.5426	5.3715	4.2484	0.6589	-0.8494	+0.1164
KAN	7.5272	4.9993	0.3303	7.3162	4.6730	0.3673	-0.2110	+0.0370

Interpretation

- Best RMSE in the 5-feature baseline: XGBoost (6.06).
- Best RMSE in the 6-feature + CI2 run: XGBoost (5.20).
- The winning model family did not change for DBCM.

3. Target-level results - BDCM

Detailed RMSE, MAE, and R2 values for every model in both ablation rounds.

Model	5f RMSE	5f MAE	5f R2	6f RMSE	6f MAE	6f R2	Delta RMSE	Delta R2
Random Forest	9.9386	7.6170	0.7820	8.8371	6.8914	0.8277	-1.1016	+0.0456
XGBoost	10.7643	8.1005	0.7443	9.8987	7.4108	0.7838	-0.8656	+0.0395
MLP	10.5631	7.8445	0.7538	8.4008	6.9316	0.8443	-2.1623	+0.0905
KAN	11.3376	8.8292	0.7164	11.4915	7.8891	0.7086	+0.1539	-0.0078

Interpretation

- Best RMSE in the 5-feature baseline: Random Forest (9.94).
- Best RMSE in the 6-feature + CI2 run: MLP (8.40).
- The winning model family changed from Random Forest to MLP after adding CI2.

4. SHAP summary across the two ablation rounds

Feature attribution changes explain where the performance gains came from.

SHAP values were computed for both formal runs. Tree-based models used TreeExplainer, while MLP and KAN used KernelExplainer on the complete-case test set. The summary below focuses on the model-target pairs that best explain the observed performance shifts.

THM4 / Random Forest

- Top features in 5-feature baseline: TOC (0.339), Bromide (0.147), UV254 (0.144)
- Top features in 6-feature + Cl2: Cl2 dose (0.214), TOC (0.189), Bromide (0.121)
- Interpretation: Cl2 becomes the strongest feature and THM4 error drops sharply.

DBCM / XGBoost

- Top features in 5-feature baseline: Bromide (0.368), Temperature (0.109), UV254 (0.106)
- Top features in 6-feature + Cl2: Bromide (0.371), Temperature (0.095), UV254 (0.078)
- Interpretation: Bromide remains dominant; Cl2 contributes only modestly.

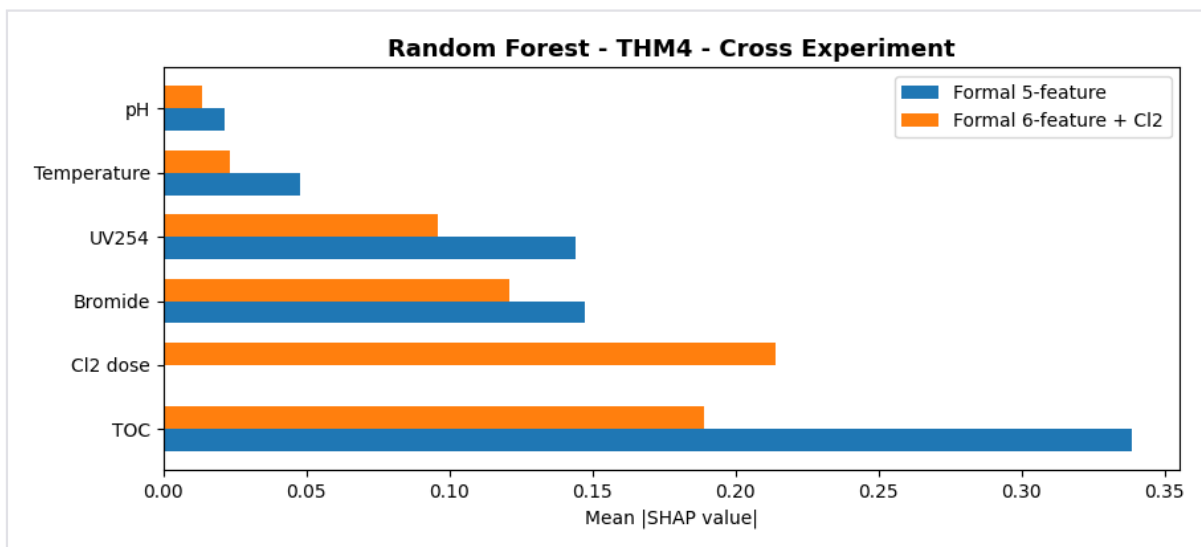
BDCM / MLP

- Top features in 5-feature baseline: TOC (0.496), Bromide (0.327), UV254 (0.135)
- Top features in 6-feature + Cl2: Bromide (0.353), Cl2 dose (0.298), TOC (0.292)
- Interpretation: Cl2 emerges as a strong second driver and helps BDCM most in MLP.
- THM4 changes the most: TOC leads in the baseline, but Cl2 becomes the top SHAP feature in all four models after it is added.
- DBCM changes the least: bromide remains the main explanatory feature in both rounds.
- BDCM keeps bromide as the main signal, but Cl2 rises to a clear second place in every model family.

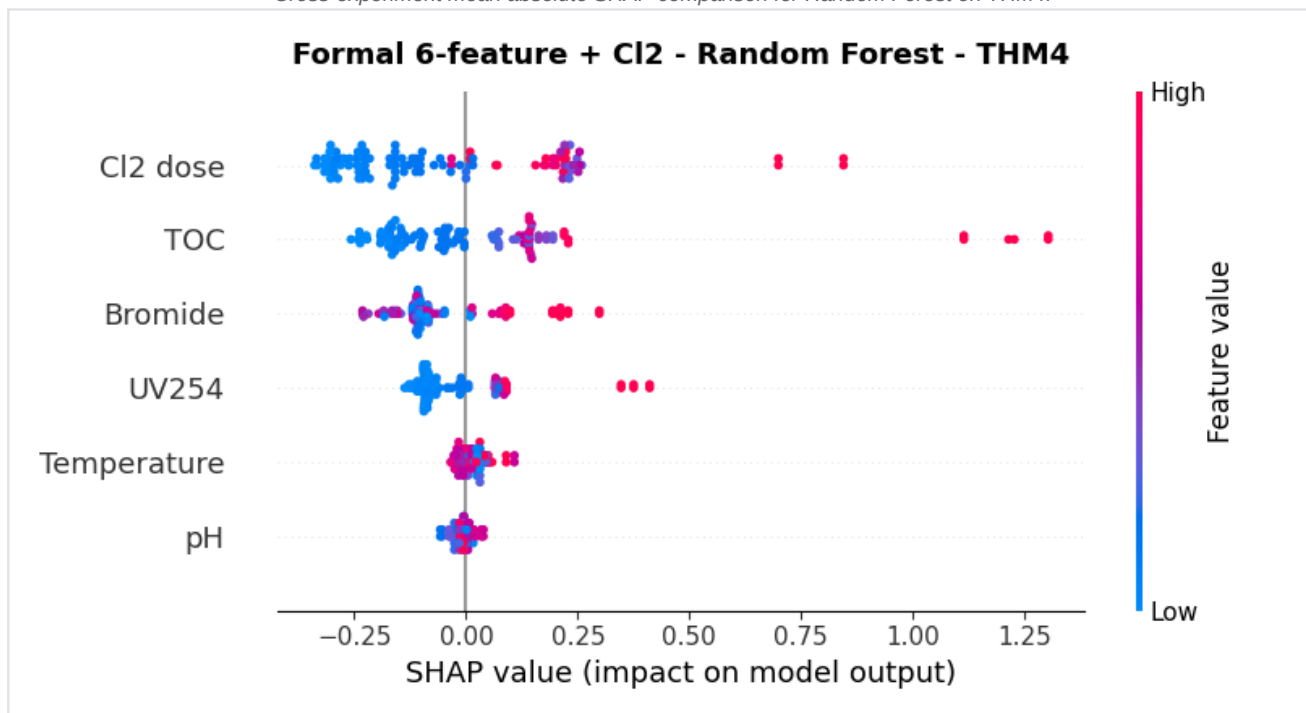
5. SHAP focus - THM4

Random Forest provides the clearest explanation for why THM4 improves after adding Cl2.

- In the baseline run, THM4 is mostly driven by TOC, with bromide and UV254 as secondary terms.
- After Cl2 is added, Cl2 dose becomes the strongest SHAP contributor and TOC remains the main secondary driver.
- This attribution shift is consistent with the large THM4 RMSE reduction from 55.43 to 41.46 in Random Forest.



Cross-experiment mean absolute SHAP comparison for Random Forest on THM4.

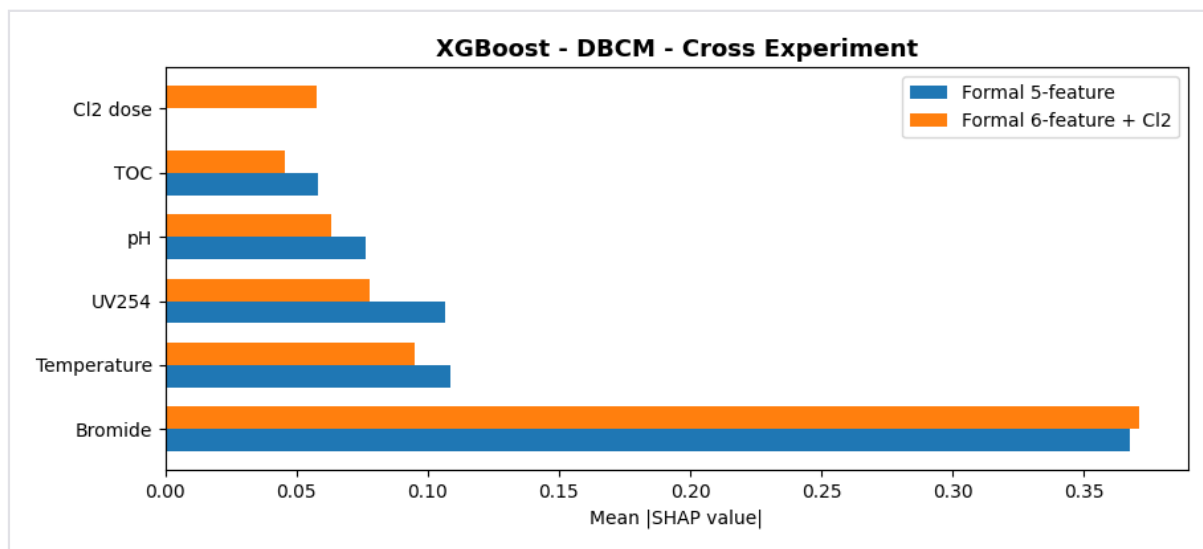


6-feature + Cl2 SHAP beeswarm for Random Forest on THM4.

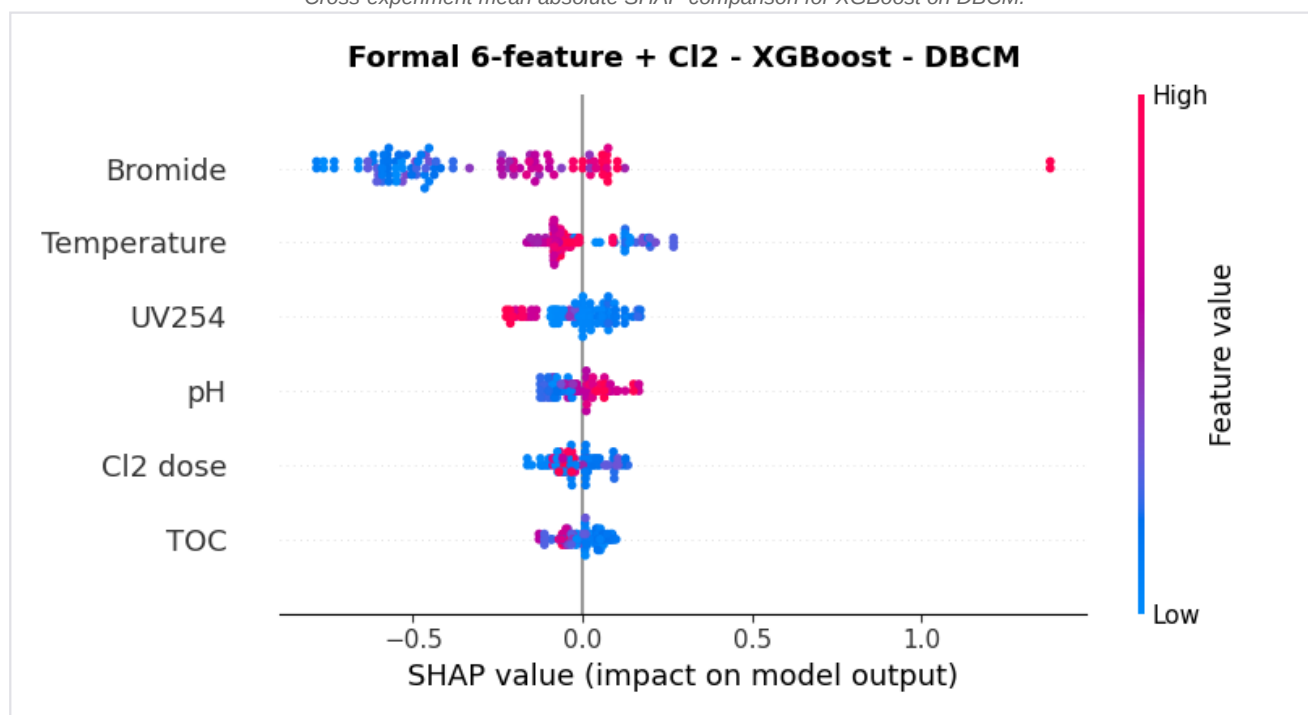
6. SHAP focus - DBCM

XGBoost remains the strongest DBCM model, and SHAP shows why Cl2 only adds limited extra information here.

- Bromide is the dominant feature in both rounds, with temperature and UV254 staying ahead of Cl2.
- Cl2 enters the model, but its SHAP magnitude remains smaller than the main bromide-driven structure.
- That pattern matches the modest but still real DBCM gain in XGBoost, from RMSE 6.06 to 5.20.



Cross-experiment mean absolute SHAP comparison for XGBoost on DBCM.

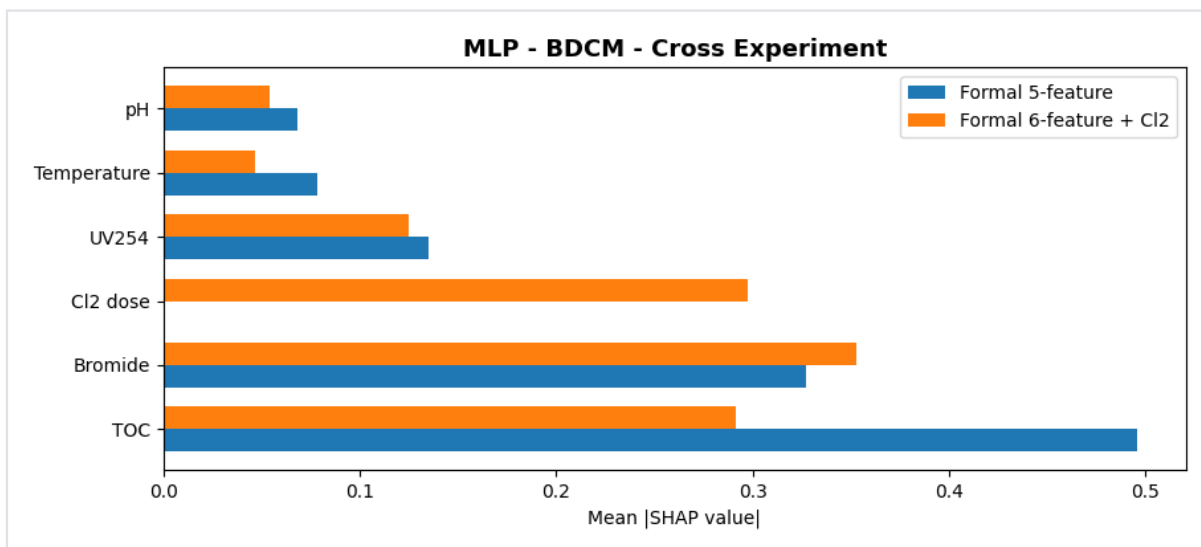


6-feature + Cl2 SHAP beeswarm for XGBoost on DBCM.

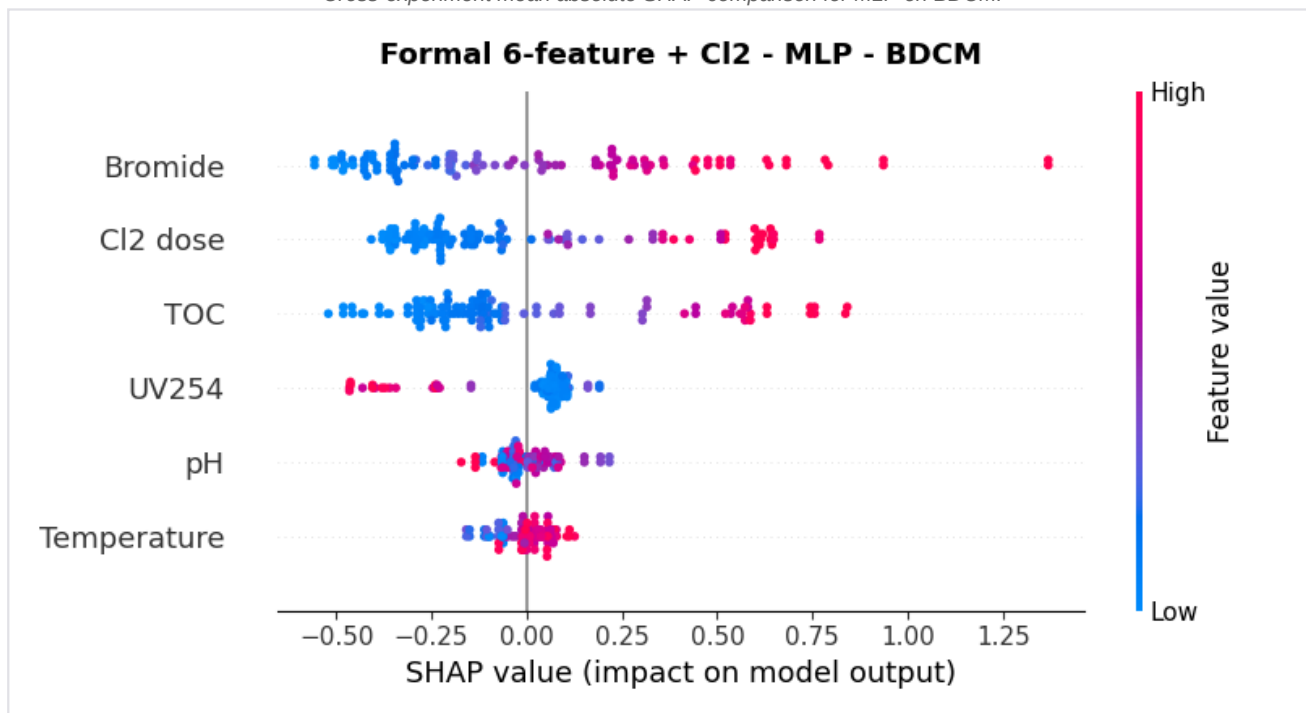
7. SHAP focus - BDCM

MLP becomes the best BDCM model in the 6-feature run, and SHAP shows that Cl2 adds a new strong signal on top of bromide.

- Bromide remains the top BDCM feature, but Cl2 rises to the second position once it is introduced.
- TOC remains important, yet the feature balance is more distributed than in the baseline run.
- This helps explain why the best BDCM model changes from Random Forest in the baseline to MLP in the 6-feature run.



Cross-experiment mean absolute SHAP comparison for MLP on BDCM.



6-feature + Cl2 SHAP beeswarm for MLP on BDCM.

8. Conclusions and recommendations

What these two formal ablation rounds tell us.

- Adding CI2 is justified: it improves three of the four model families and produces the strongest benefit for THM4.
- THM4 is the target most directly reshaped by CI2, both in predictive accuracy and in SHAP attribution structure.
- DBCM remains primarily bromide-driven, so CI2 should be treated as a supporting rather than dominant signal for that target.
- BDCM benefits from CI2 as an important secondary driver, especially in the MLP model where it helps change the winning model family.
- For future reporting, Random Forest is the clearest headline model for THM4, XGBoost for DBCM, and MLP for BDCM.

All figures in this report were generated from the formal dataset1 checkpoints and the SHAP analysis outputs in the repository. If the experiments are rerun, this PDF can be regenerated directly from the updated artifacts.